Project Proposal

VLSI CAD (EC440)

RTL TO GDSII:
ETHERNET MAC

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OBJECTIVE

Primary Goal: To implement a complete, automated RTL-to-GDSII physical design flow for an IEEE 802.3 compliant Ethernet Media Access Control (MAC) core using open-source EDA tools.

Deliverable	Description
GDSII Layout	Final manufacturable layout file compatible with standard fabrication processes
Gate-Level Netlist	Optimized digital logic representation mapped to the standard cell library
Physical Verification	Sign-off reports confirming zero Design Rule Checking (DRC) violations and passing Layout vs. Schematic (LVS) checks
Performance Reports	Detailed analysis of timing constraints, area utilization, and power consumption

LITERATURE **SURVEY**

ETHERNET MAC CORE

- Ethernet MAC IP cores implement IEEE 802.3 / 802.3u Layer-2 framing at 10/100 Mbps and typically sit between a system bus on the host side and an external PHY on the line side.
- The OpenCores Ethernet IP Core provides key MAC functions such as CRC generation/checking, preamble handling, padding, half-duplex collision handling (CSMA/CD), and full-duplex flow control using PAUSE frames (IEEE 802.3x).
- Its architecture is commonly described as a set of blocks including TX MAC, RX MAC, a MAC control block (flow control), an MIIM management block (MDIO/MDC), and a host interface using WISHBONE.

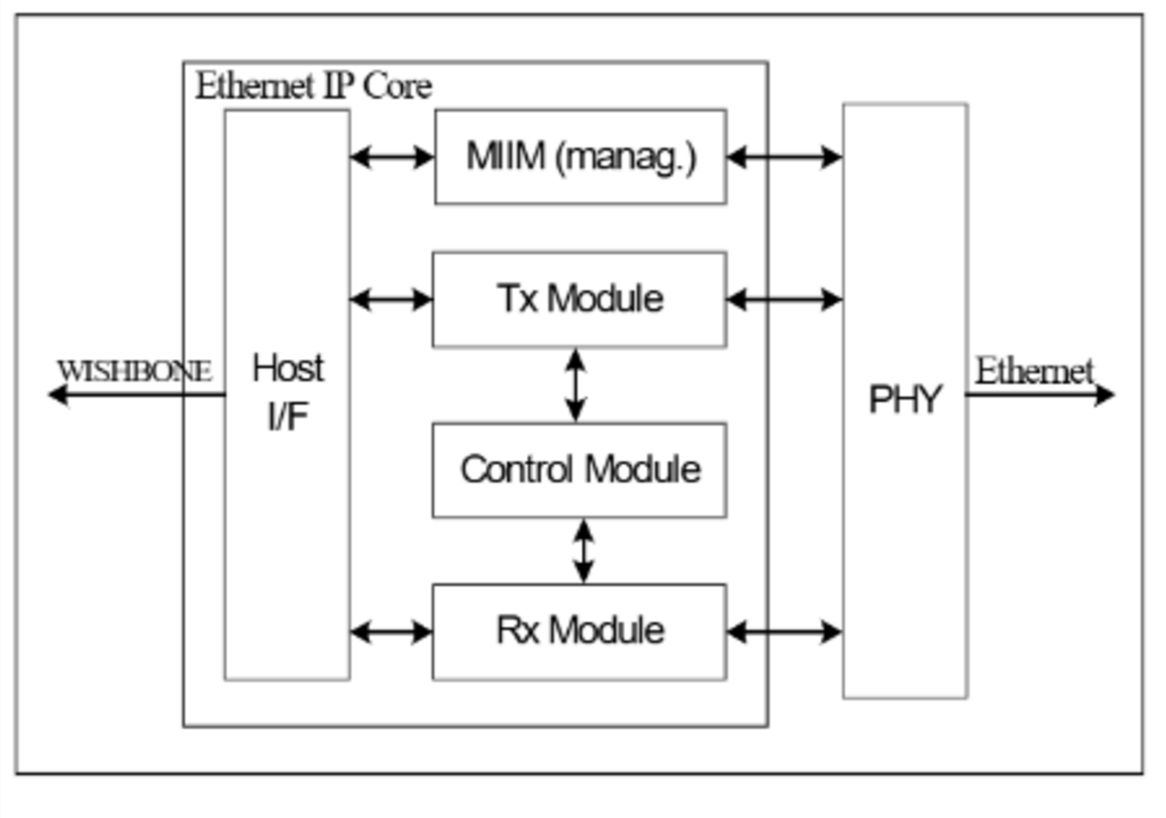

Figure 1: Core Architecture Overview

Technology	Area	Silicon Area	Speed	Power Consumption
Xilinx Virtex	2600 slices	100000 gates	60 MHz (25 needed)	TBD

Figure 2: Estimated metrics

LITERATURE **SURVEY**

1. FUNCTIONAL MODULES

The core consists of seven primary modules, each handling specific Layer-2 responsibilities:

Module Name	Primary Function
Host Interface	CPU/Memory Communication
TX MAC	Frame Transmission
RX MAC	Frame Reception
MAC Control	Flow Control
MII Management	PHY Configuration
Registers	Configuration & Status
Status Module	Error Reporting

LITERATURE **SURVEY**

2. ARCHITECTURE DIAGRAM

The diagram below illustrates the data and control flow between these modules:

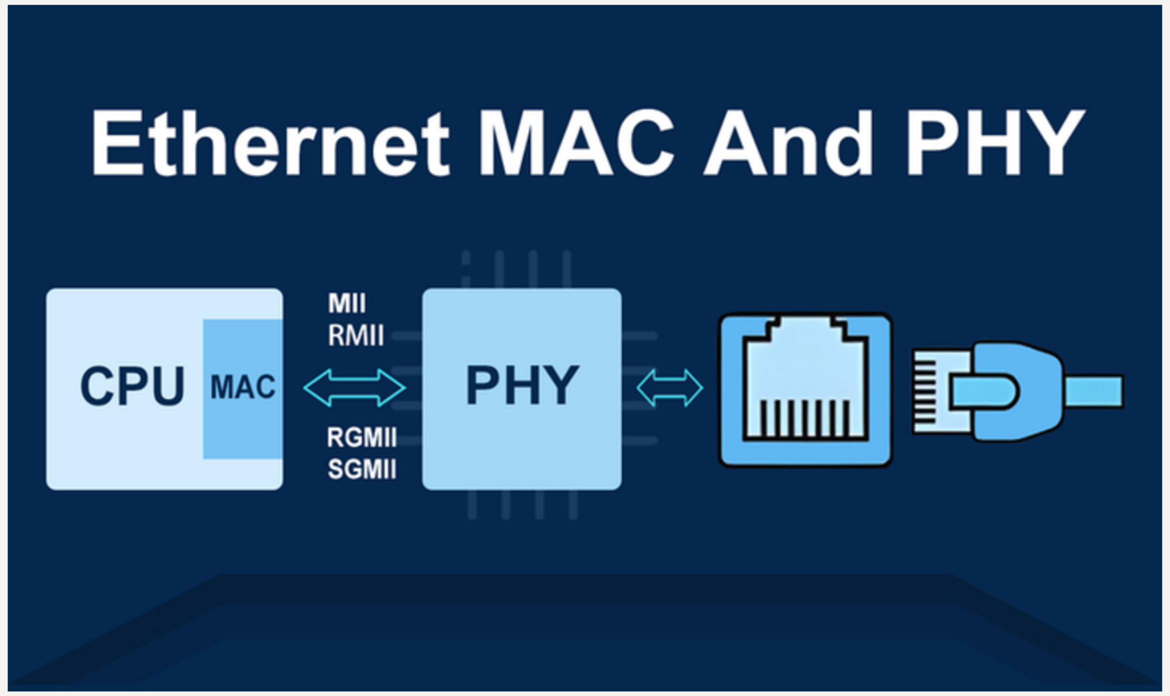

Figure 3: Overview

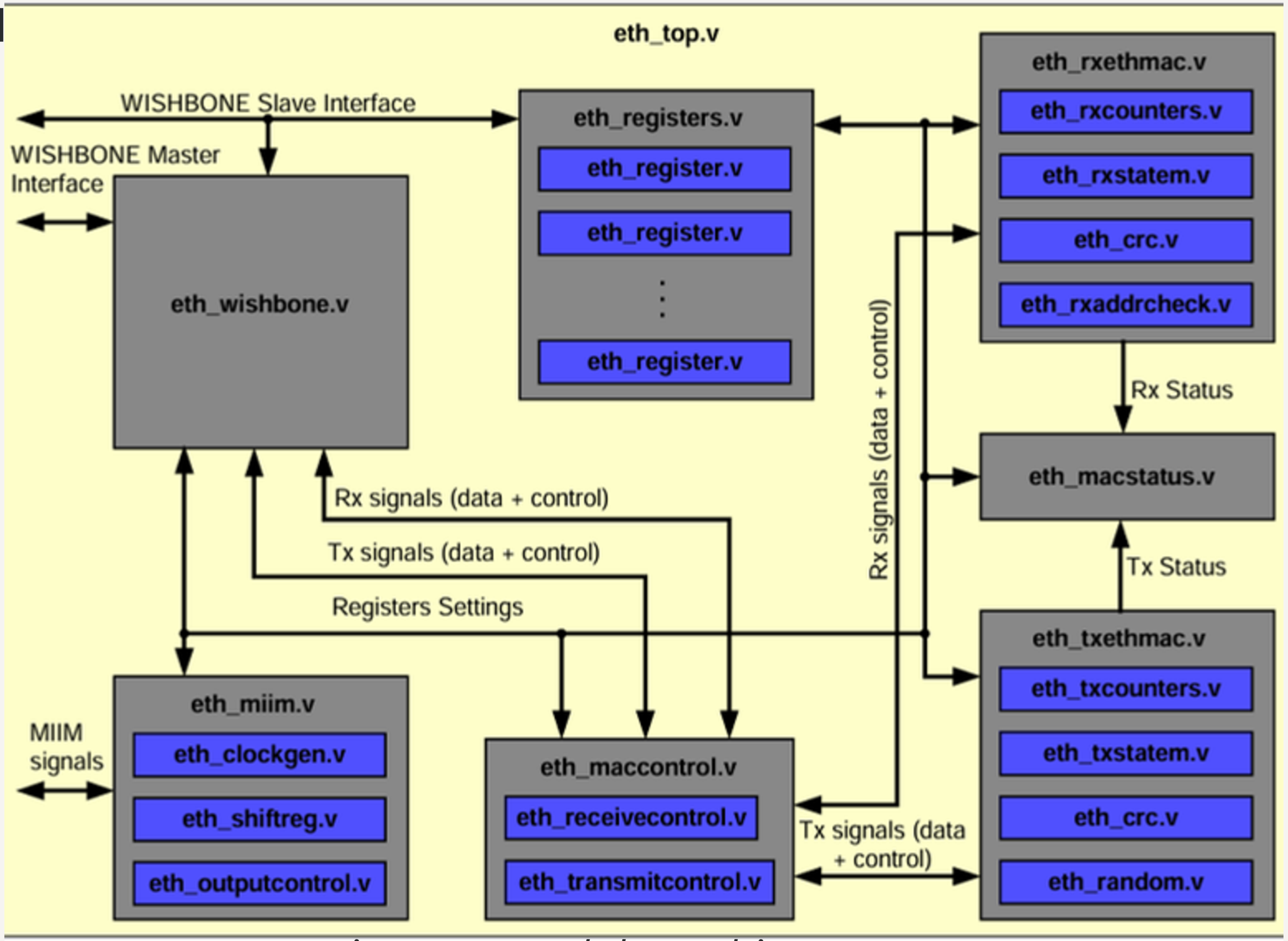

Figure 4: Module Architecture

LITERATURE **SURVEY**

ETHERNET FRAME

Figure 5: Ethernet Frame

- **Preamble**: this is a 7-byte pattern of ones and zeroes and is used for synchronization.
- **SFD**: the "start frame delimiter' marks the end of the preamble and tells the receiver that the next fields will be the actual Ethernet frame, starting with the destination field.
- **Destination**: this is the destination MAC address of the receiver.
- **Source:** the source MAC address of the device that sent the frame.
- **Type**: this tells us what is carried inside the Ethernet frame. An IPv4 packet, IPv6 packet or something else.
- **Data**: this carries the actual data that we are trying to transmit, for example an IPv4 packet.
- **FCS**: the frame check sequence helps the receiver to figure out if the frame is correct or corrupt.

LITERATURE **SURVEY**

ESTIMATED WRAPPING OUTPUT

Interface	Signal Group	Pin Count	Purpose
Clock & Reset	CLK, RST	2	System clock and reset
WISHBONE Host Bus	Address	8	256 register locations
	Data (Bidirectional)	32	32-bit data bus
	Control (WE, ACK, etc.)	4	Write enable, acknowledge,
	Subtotal	**44**	
MII TX Interface	MTxD[3:0], MTxEn, MTxErr	6	TX data nibble, enable, error
MII RX Interface	MRxD[3:0], MRxDV, MRxErr	6	RX data nibble, valid, error
MII Management	MDIO, MDC, Mdi, Mdo, MdoEn	5	PHY management interface
	Subtotal	**17**	
Power & Ground	VDD, GND	4 to 8	Power distribution (multiple
	Subtotal	**6**	(typical for digital core)
Interrupt	IRQ	1	Interrupt output to CPU
Total I/O Pins		**~68–72**	Depending on power pin

LITERATURE **SURVEY**

OPENLANE

OpenLane is an automated, open-source RTL-to-GDSII flow developed to support the SkyWater 130nm PDK. It integrates various EDA tools into a unified pipeline using custom scripts and abstractions.

- Role: Acts as the primary framework ("flow controller") for this project, automating the steps from RTL synthesis to final layout generation.
- Key Feature: Enables "push-button" flow execution while allowing deep configuration of each stage (Synthesis, Floorplan, Placement, CTS, Routing) via a config.tcl file.

YOSYS

Yosys is a framework for Verilog RTL synthesis.

- Role: It takes the Ethernet MAC Verilog source code and converts it into a gate-level netlist (a list of connected logic gates like AND, OR, D-FlipFlops) mapped to a standard cell library.
- Key Feature: Supports advanced mapping to technology libraries (Liberty format) and logic optimization to minimize area and delay.

LITERATURE **SURVEY**

OPENROAD

OpenROAD is an integrated physical design tool that aims to achieve 24-hour, no-human-in-the-loop layout generation. It replaces older distinct tools by combining multiple engines into one application.

- Components Used:
  - RePlAce and OpenDP : For floorplanning and placement of cells.
  - TritonCTS: For synthesizing the clock tree network.
  - TritonRoute: For detailed routing of metal interconnects.
  - OpenSTA: For static timing analysis (timing sign-off).
- Role: Handles the core physical implementation steps: Floorplanning, Placement, CTS, and Routing.

MAGIC, NETGEN, AND KLAYOUT

These tools handle the physical verification and "sign-off" stages, ensuring the layout is manufacturable and correct.

- Magic: A VLSI layout tool used for Design Rule Checking (DRC), ensuring geometry meets foundry constraints, and for Antenna checking.
- Netgen: A netlist comparison tool used for LVS (Layout vs. Schematic), verifying that the physical layout's connectivity matches the synthesized netlist.
- KLayout: A high-performance GDSII viewer used for visual inspection of the final layout file.

PROCESS OF **IMPLEMENTATION**

The RTL-to-GDSII implementation flow is executed sequentially through the following stages:

Stage	Process Name	Tool(s) Used
1	**RTL Synthesis**	**Yosys**
2	**Floorplanning**	**OpenROAD**
3	**Placement**	**OpenROAD**
4	**Clock Tree Synthesis (CTS)**	**OpenROAD**
5	**Routing**	**OpenROAD**
6	**Physical Verification**	**Magic** (DRC) **Netgen** (LVS)
7	**GDSII Generation**	**Magic / KLayout**

RESOURCES **REQUIRED**

The following hardware, software, and design data are necessary to execute the complete RTL-to-GDSII flow for the Ethernet MAC core:

SOFTWARE & TOOLS

Software	Role in Flow
OpenLane	Automated flow controller and integration framework
Yosys	RTL synthesis and logic mapping
OpenROAD	Floorplanning, placement, CTS, and routing
Magic	Layout viewing, DRC checking, and GDS extraction
Netgen	LVS (Layout vs. Schematic) verification
KLayout	GDSII viewer and final layout inspection
Docker	Containerization to run the OpenLane environment consistently

PROJECT **TIMELINE**

The implementation of the RTL-to-GDSII flow is estimated to take approximately 8 weeks. The schedule is broken down into four main phases:

Week	Phase	Key Activities & Milestones
Week 1-2	**Phase 1: Setup & Synthesis**	- Environment setup - RTL code analysis and linting. -Wrapper creation - **Milestone:** Successful Synthesis (Gate-level Netlist generation).
Week 3-4	**Phase 2: Physical Design**	- Floorplanning (Core area, I/O pin placement, Power planning). - Global and Detailed Placement. - Clock Tree Synthesis (CTS). - **Milestone:** Placed Design with CTS complete.
Week 5-6	**Phase 3: Routing & Optimization**	- Global and Detailed Routing. - Handling congestion and routing violations. - Timing optimization (Setup/Hold fix). - **Milestone:** Fully Routed Design (DRC Clean). **Midsems here**
Week 7	**Phase 4: Verification (Sign-off)**	- Running Magic DRC (Design Rule Check). - Running Netgen LVS (Layout vs. Schematic). - Antenna violation checks. - **Milestone:** Verified Layout (LVS Match, Zero DRC errors).
Week 8	**Phase 5: Final Delivery**	- GDSII Stream-out generation. - Final report compilation (Area, Power, Timing). - Project documentation and submission. - **Milestone:** Final Project Submission.

REFERENCES

- *I. Mohor, "Ethernet IP Core Specification," OpenCores.org, Rev. 1.19, Nov. 2002. [Online]. Available: https://github.com/freecores/ethmac*
- *I. Mohor, "Ethernet IP Core Design Document," OpenCores.org, Rev. 0.4, Oct. 2002.*
- *IEEE, "IEEE 802.3-2022 Standard for Ethernet," 2022.*
- *Yosys Open Synthesis Suite. [Online]. Available: https://yosyshq.readthedocs.io/*
- *OpenROAD Project. [Online]. Available: https://openroad.readthedocs.io/*
- *OpenLane Documentation. [Online]. Available: https://armleo-openlane.readthedocs.io/*
- *Magic VLSI Design System. [Online]. Available: http://opencircuitdesign.com/magic/*
- *NetGen Tool. [Online]. Available: http://opencircuitdesign.com/netgen/*
- *KLayout Project. [Online]. Available: https://www.klayout.de/*
- *NetworkLessons, "Introduction to Ethernet." [Online]. Available: https://networklessons.com/switching/introduction-to-ethernet*
- *N. Raphaely, et al., "OpenLane: The Open-Source Digital ASIC Implementation Flow," 2020.*

THANKYOU